

Modeling and Simulation of Nonlinear Loads in the IEEE 34-Bus Feeder Using OpenDSS

Soumya Ranjan Das
Department of Electrical Engineering, IIT (ISM) Dhanbad

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Abstract

This project focuses on simulating linear, nonlinear, switching load behavior and step load variations in the IEEE 34-bus distribution feeder using OpenDSS. Harmonic distortion, voltage response, and time-series load changes are modeled using custom spectrum definitions and simulation controls. The goal is to analyze power quality metrics and voltage performance under distorted and dynamic loading conditions.

1 Introduction

Nonlinear, switching loads such as power electronic devices and industrial equipment inject harmonic currents into the power system, affecting voltage quality.

I have made the IEEE34 bus Simulation in open dss and simulate it with first some linear and constant loads connected, then switching and constant loads then all constant loads and finally mixing all these.

2 Bus Diagram

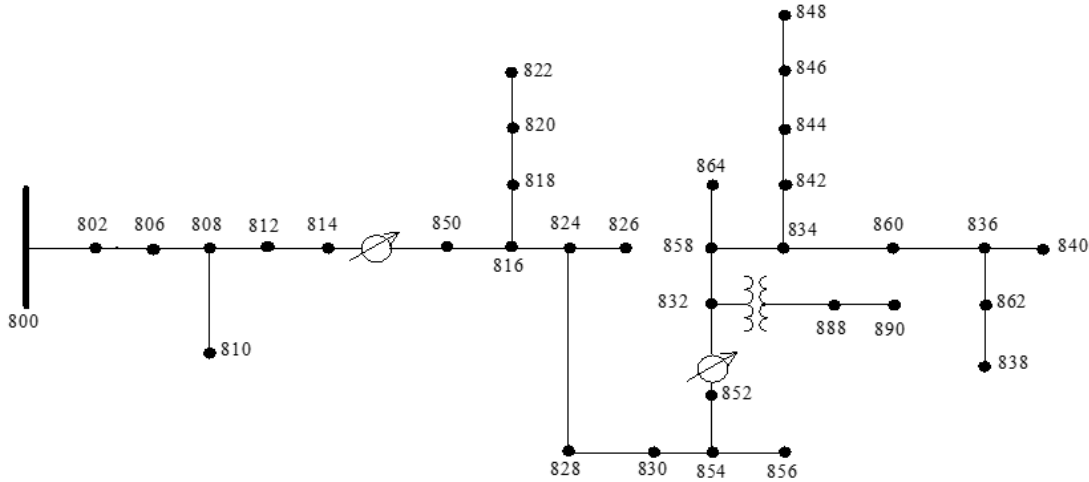


Figure 1: IEEE 34-Bus Feeder Topology

3 System Description

The IEEE 34-Bus distribution feeder is used as the base system. **This feeder is an actual feeder located in Arizona back in year 1992**, with a nominal voltage of 24.9 kV. It is characterized by long and lightly loaded, two in-line regulators, an in-line transformer for short 4.16 kV section, unbalanced loading, and shunt capacitors. The simulation includes:

- Voltage regulators
- Capacitor banks
- Transformer banks with single-phase control

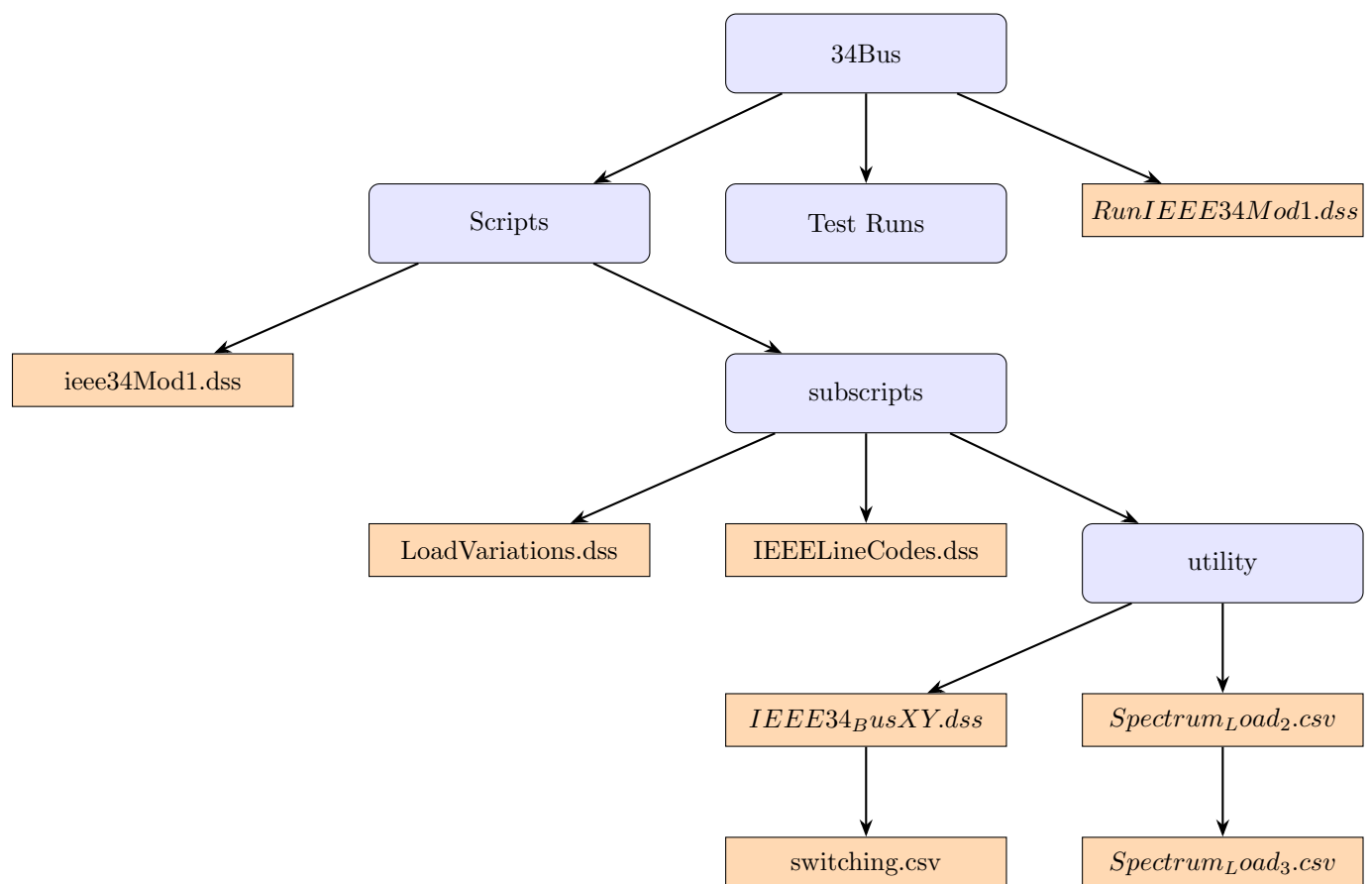
I have made the IEEE34 bus Simulation in open dss first with 20 constant loads connected , then 20 linear loads connected, 20 Switching loads and at last 20 loads with combination of these 3 types of loads.

I am doing a time series simulation of step size 24 hrs.

So when the switching loads are connected the loads step up by 30 - 40 percent at start and reduces by 20 - 30 percent after 12 hours and the cycle continues.

There are total 68 loads mix of 1 phase and 3 phase.

4 File Architecture



File Architecture

5 Working Plan

